「Def. A map

$$\exp : \mathbb{C} \to \mathbb{C} : z \mapsto \sum_{n=0}^{\infty} \frac{z^n}{n!}$$

is called the exponential. Observe the following ; given a real-value y,

$$e^{iy} = \sum_{n=0}^{\infty} \frac{(iy)^n}{n!} = \sum_{n=0}^{\infty} (-1)^n \cdot \frac{y^{2n}}{(2n)!} + i \cdot \left(\sum_{n=0}^{\infty} (-1)^n \cdot \frac{y^{2n+1}}{(2n+1)!} \right)$$

$$= \cos y + i \cdot \sin y$$